Midterm Project Proposal

Used Car Price Prediction — Dataset Description & Project Plan

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1. Dataset Description

The dataset used in this project is the Used Car Listings dataset sourced from Kaggle. It contains 9,582 records of second-hand vehicle listings from India, scraped from an online automobile marketplace. Each record represents a single listing with 11 raw features covering brand, mileage, transmission, fuel type, ownership history, and the listed asking price.

Source URL: https://www.kaggle.com/datasets/used-car-listings

Total records: 9,582 rows | 11 columns (raw)

Target variable: AskPrice (listed price in Indian Rupees)

Column	Type	Description
`Brand`	Categorical	Car manufacturer (e.g., Maruti Suzuki, Toyota, BMW, Hyundai)
`model`	Categorical	Vehicle model name — dropped during prep (high cardinality)
`Year`	Numeric	Year manufactured — dropped (collinear with Age)
`Age`	Numeric	Age in years; key predictor — newer cars command higher prices
`kmDriven`	Numeric*	Total km driven (raw: string with commas and ' km' suffix)
`Transmission`	Categorical	Manual or Automatic — Automatic typically priced higher
`Owner`	Categorical	First or second owner — first-owner cars attract a premium
`FuelType`	Categorical	Petrol, Diesel, or Hybrid/CNG
`PostedDate`	String	Listing month-year — not a car attribute; dropped
`AskPrice`	Numeric*	Target variable: listed price in Rupees (raw: string with Rs symbol)

** kmDriven and AskPrice are raw strings in the source file and require parsing before use.*

2. What We Will Accomplish

The goal is to build a supervised regression model that accurately predicts the asking price of a used car from its attributes. This has direct commercial value: accurate valuations help buyers avoid overpaying and help sellers price competitively. The project encompasses full data preparation and comparative model evaluation.

Data Preparation Tasks:

- **Parse raw strings:** Convert kmDriven (e.g., '98,000 km') and AskPrice (e.g., 'Rs 1,95,000') from formatted strings to numeric floats.
- **Handle missing values:** Impute the 47 null values in kmDriven using the median calculated on training data only.
- **Remove duplicates:** Drop 724 exact duplicate rows present in the raw dataset.
- **Feature engineering:** Drop Year (collinear with Age), AdditionInfo and PostedDate (low signal), and model (too many categories).
- **Brand filtering:** Retain only the top 16 brands by frequency; rows with rare brands are removed to reduce noise.
- **Encoding:** Apply one-hot encoding to Brand, Transmission, Owner, and FuelType, producing integer 0/1 columns.
- **Standardization:** Z-score scale Age, kmDriven, and AskPrice (mean=0, std=1) using a scaler fitted on training data only.
- **Train/Test split:** Hold out 30% of data as an untouched test set before any cleaning or scaling is applied.

Modelling Plan:

Three regression models will be trained and compared on the cleaned training set, then evaluated on the held-out test set using MAE, RMSE, and R^2:

Model	Strengths	Primary Metric
Linear Regression	Interpretable baseline; coefficient ranking of features	R^2, MAE
Random Forest	Non-linear patterns; robust to outliers; feature importance	MAE, RMSE
XGBoost	Highest accuracy on tabular data; handles missing values	RMSE, R^2

Dataset: 9,582 raw rows | Cleaned training set: 5,973 rows | 26 features after preparation | Spring 2026